

Prepositioning and distributing relief items in humanitarian logistics with uncertain parameters

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ABSTRACT

Humans tackle natural disasters all over the world. Humanitarian supply chain plays an important role to mitigate damages occurred after a disaster. This research formulates a multi-objective Mixed-Integer Non-Linear programming (MINLP) with uncertain parameters considering Relief Centers (RC), Demand Points (DP) in affected areas, transportation methods to deliver Relief Items (RI) and different types of RIs namely, perishable and imperishable. In pre-disaster stage, location and number of RCs with their prepositioned inventory level are determined. After disaster strikes, based on a distribution plan the amount of RIs that should be transported to DPs and number of needed vehicles are determined. The objective functions minimize the total distance traveled by RIs, total costs (including RC establish cost, inventory cost, fixed cost for each vehicle type and acquisition cost for RIs), maximum traveling time between RCs and DPs and number of perished items respectively. The proposed model is solved by GAMS software for small size test problems and Grasshopper Optimization Algorithm (GOA) as a meta-heuristic approach for large size problems. Numerical and computational results are provided to prove the efficiency and feasibility of the presented model. Finally, the developed model is implemented to Iran's flood in 2019 as a case study.

1. Introduction

Natural disasters like earthquake and flood have a very devastating effect on the basic infrastructure of the countries like physical structures, economic, society, etc. and strongly influence the suffered people in crowded regions [1,2]. From 2003 to 2012, about 106,654 people are annually dead, around 216 million people were affected by disasters according to reports, and about \$157 billion were reported to be damaged economically [3]. Furthermore, from the beginning of the twentieth century, there are more than eight million deaths because of technological and natural disasters in the world [4]. Two devastating earthquakes that occurred in April 2015 and May 2015 in Nepal are two examples of recent disasters. About 8000 people were killed and more than 2.8 million people were needy for humanitarian aid [5]. To mitigate these effects on humans, a disaster management approach based on humanitarian relief logistics network should be implemented [6].

Disasters are usually unpredictable, and it causes many casualties. Consequently, humanitarian logistics (HL) plays a key role in improving relief operation in different ways such as cost, time and quality [7]. HL includes various activities such as disaster preparedness and planning,

transportation, procurement and warehousing [8]. These activities can be categorized into four groups, including preparedness, mitigation, response, and recovery. Preparedness and mitigation are those activities which should be performed before the disaster (pre-disaster phase) like pre-positioning RIs. Response and recovery activities are those activities which should be performed after the disaster (post-disaster phase) like distribution of RIs or evacuating suffered people [9]. Transporting resources such as relief commodities and equipment to the disaster area with its all predecessor and successor operations can contribute to about 80% of the total relief costs [10]. Preparedness phase aims to get prepared for any plausible disasters while the response phase is related to those tasks hindering disaster incidences [11]. Pre-positioning leads to faster response and more effective procurement and distribution; however, it needs further investment before the disaster occurs [12].

This research contributes to the literature in various ways. First, we focus on developing a mathematical model for pre-positioning several RCs and determining inventory level in each of them. Many researches have only paid attention to imperishable commodities in RCs (e.g., [13–15]) though we consider perishable items as well. Besides, Perishable and imperishable RIs should be transported with different type of

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vehicles because perishable RIs such as medicines or foods need to be in special conditions; for example, they need colder temperature or air circulation system to retain their quality. Secondly, we considered different type of costs, transportation methods with different capacities and a new multi-objective function. Besides, by regarding time window for loading and unloading RIs on vehicles, we surveyed on two main concerns in HL, namely, minimizing distance traveled by RIs and maximum travel time. Finally, we used a meta-heuristic algorithm to solve the proposed model.

To sum up, in this research, we are looking forward to answering these questions:

- How many and where RCs should be established considering constraints such as cost issues and accessibility?
- How should each RC be assigned to a DP concerning perishable items that may decay before they deliver)?
- How many RIs should be held at each RC considering all type of costs as uncertain parameters (such as acquiring cost, holding cost, etc.)?
- How many vehicles are needed to deliver RIs?

The rest of this paper is organized as follows. In the next section, we present the related literature and develop our hypotheses. In the third section, the problem is defined and the mathematical model is presented. Solution method, model validation and numerical results are provided in the fourth section. A case study is applied to the proposed model in section five. Analysis of model parameters are surveyed in section six. We provided some useful managerial insights in section eight and finally, conclusion and future research directions are discussed in the ninth section.

2. Literature review and hypothesis development

Researchers have reviewed the HLs area from various standpoints. Several reviews have been performed by researchers recently such as comprehensive review on HL [16], general review on humanitarian relief chains [11] and review on inventory management in humanitarian supply chains [17]. The most common objectives function in pre-positioning literature are minimizing transportation costs, RC establishing costs [6,18,19], RI costs [20], and maximizing responsibility and coverage [5,21–25]. Campbell and Jones [15] proposed a model for deciding where a supply center should be located before a disaster. Their model determines the optimal inventory of RIs and tries to minimize the total costs related to delivery to demand center from a supply point. Rawls and Turnquist [26] develop a model for emergency response that considers the location and amount of different types of emergency items to be pre-positioned. Their paper represents a two-stage stochastic mixed-integer program considering uncertainty in demand for relief commodities and in transportation network is availability after the disaster. Ukkusuri and Yushimito [27] claim that one of the strategies that can increase post-disaster response is inventory prepositioning because it can reduce the delivery time. They presented a model to minimize both fixed costs and routing costs. Another study performed by Maharjan and Hanaoka [5] tried to determine the optimal number and locations of RCs in different places of Nepal that aim to maximize the coverage. They applied a branch and bound method to solve the model. Tofghi et al. [28] introduced a two-echelon HL network with central warehouses and local distribution center before and after the disaster phase. They developed the proposed model with a two-stage scenario-based possibilistic-stochastic programming approach. Döyen et al. [24] proposed a two-stage stochastic mixed-integer linear programming model for the humanitarian supply chain for both pre and post-disaster phase. The aim of their model is to minimize costs relating to facility location, inventory, transportation and shortage. They solved the model by a heuristic method that is based on Lagrangean relaxation. Another study which considers pre-disaster and post-disaster inventory level and procurement have been carried

out by Hu et al. [29]. They applied a scenario-based approach to demonstrate the uncertain demand and considered uncertainty about disaster type. Li et al. [30] focused on developing a model to maximize cooperative covering and considering the budget to maximize benefits to affected people in the disaster area. They assumed demand as an uncertain parameter and used financial efficiency and demand coverage as two indicators to evaluate humanitarian relief chain management. Bozorgi-Amiri et al. [12] represented a multi-objective robust stochastic programming approach for HLs. They considered demand, cost of procurement and transportation as uncertain parameters. The proposed model tries to minimize the summation of the expected value and the variance of the total cost. Furthermore, the model aims to maximize the satisfaction level in affected regions by minimizing the summation of the maximum shortage. Khayal et al. [18] proposed a network flow model for selecting impermanent distribution centers and allocating resources dynamically in various periods. Rezaei-Malek et al. [31] presented a multi-objective, two-stage stochastic, non-linear, and mixed-integer model to optimize three objectives (i.e., efficiency, efficacy, and imbalance) in a humanitarian relief chain. Kutanoglu and Mahajan [14] proposed an integer non-linear mathematical model in order to minimize the stock cost for a two-echelon logistics system which consists of a central warehouse with infinite capacity and some local warehouses. Moreover, local warehouses are allowed to share their inventory to increase the service level. Ozkapici et al. [32] introduced a maritime transportation model for HLs in the aftermath of an earthquake in Istanbul. They considered the vulnerability on the transportation time of commodities. Minimizing the demand-weighted transportation time of relief goods was the objective of the presented model. Lin et al. [19] presented a model to minimize the four types of operational relief cost. They presumed a central depot and several locations for temporary depots near the disaster region and resources and vehicles to improve the HL efficiency. Moreover, they developed a two-phase heuristic method to solve their model. Ukkusuri and Yushimito [27] modeled the pre-positioning supply centers for disasters as a facility location problem and considered routing for vehicles and probability for transportation network disruptions. The proposed integer-programming model tries to use the most reliable route. Afshar and Haghani [33] developed a general mathematical model that controls the RI flow from supply centers until demand centers. Their model allows for a centralized plan. It considers many details i.e. vehicle routing (VRP), planning for delivery and pick up, optimal locations for impermanent facilities and capacity limitation for the transportation system and each facility center. Bemley et al. [34] proposed a two-staged model for pre-positioning. They considered warehouses for humans and material assets before the disaster and distribute them after the disaster. They also introduced a response ratio to demonstrate the level of restoration that can be accomplished. Galindo and Batta [6] offered an integer-programming model for pre-positioning before the disaster. Demolition of supply centers is allowed and demand is assumed stochastic. Mete and Zabinsky [35] present a stochastic optimization method for storage and distribution of medical commodities when a disaster occurs. They developed their model with a wide range of possible disaster types and depending on which disaster occurs different scenarios may happen. Wong et al. [36] considered a two-location inventory system which in emergency cases lateral shipments is allowed. The inventory contains multi-item repairable spare parts. Baskaya et al. [10] studied on a real-life earthquake in Istanbul. They introduced three models; direct shipment model, lateral transshipment model, and maritime lateral transshipment that aims to minimize average distance each RI travels. They also claimed that lateral transshipment and maritime lateral transshipment have better results because of lateral transshipment option and affected people obtain faster and better service level. Rodríguez-Espíndola et al. [37] introduced a system containing optimization-for determining relief facilities location, product prepositioning, resource allocation and distribution plan- and geographical information systems –to avoid choosing a floodable relief facility-for disaster preparedness. Rodríguez-Espíndola

and Gaytán [38] proposed a two-stage method tackling with a flood; first, evaluate the flood situation and then a multi-objective model for minimizing total costs (acquisition, shipping and facility preparation cost) and total distance traveled. Ahmadi et al. [39] suggested a model with different type of vehicles, standard relief time and considering failure in network and random travel time to minimize total distribution time, costs relating to unsatisfied demand penalty, and opening local distribution centers. They developed their model with a two-stage stochastic program and solved it by GAMS. Condeixa et al. [40] proposed a model that uses conditional value at risk for pre-positioning, location and distribution before the disaster occurs. Then a post-optimization analysis is performed to measure the model performance. Dufour et al. [41] surveyed on potential advantages of adding local distribution centers to the United Nations humanitarian response depot in Kampala and Uganda. They declared that their proposed solution would be more cost-efficient. Hu et al. [29] presented a two-stage stochastic programming model with several uncertain parameters –demand, disaster type and occurrence location-dealing with inventory level before disaster and supplier selection and procurement quantity. They considered three characteristics, namely lead time discount, return price and equity for their model and conducted conditional value at risk to evaluate risk at different confidence levels. Manopiniwes and Irohara [42] developed a stochastic mixed-integer model for facility and inventory pre-positioning, evacuation and vehicle planning in both pre- and post-disaster activities. The objective function of the proposed model tries to minimize costs of establishing distribution centers, shipping, and response time with to approaches, cost-based and equity-based. Rahmani et al. [43] proposed a robust humanitarian relief chain model under uncertainty related to the epicenters of the earthquake and their magnitude. In addition, they considered that public centers could be used to establish a RC. Yahyaei and Bozorgi-Amiri [44] addressed the problem of designing a relief network considering interval uncertainty and RC disruption. They introduced a mixed-integer linear model to minimize the total cost of transportation and establishment of the unreliable distribution center and supportive distribution center.

Based on reviewing recent literature, it appears to be some research gaps in HLs. First, in most of the research, all RIs are considered imperishable, whereas medicine and food are two important items which may expire or decay. Second, in practical situations, especially after the disaster, uncertainty level increases in all aspects; however, most of the papers consider only demand as uncertain parameter. Finally, vehicles play an important role to deliver RIs; however,

researchers have paid less attention to determine optimal number of vehicles and their loading and unloading time ... Table 1 illustrates some related literature review of HLs.

Several hypotheses made in our research, that will be proved either they are correct or incorrect in sensitivity analysis section, are as follows:

- The severity of disaster has a direct impact on operational costs and distribution plan.
- By increasing distances and no limitation for maximum facilities to open, more facilities will be opened
- Since the capacity of RCs are unlimited, shortage for RIs is not acceptable except perishable commodities that may decay.
- By neglecting cost issues, RCs will be established in all potential locations.

3. Problem description and mathematical formulation

3.1. Problem description

After the occurrence of any disaster, affected people need to be protected as soon as possible. Because of the unpredictability nature of the natural disaster, in this research, we propose a comprehensive mathematical model for a two-echelon humanitarian relief chain before the disaster and a distribution plan after it. RCs are in the upper echelon and DPs are in the lower echelon. We have considered several potential locations for establishing RCs before the disaster in order to send RIs to affected areas in a shorter time. After establishing RCs and allocating DPs, the inventory level in each RC and number of needed vehicles are determined according to the population who live in that region. There is a limitation in building RCs and only a specific number of facilities can be established because establishing new RCs are very costly; however, in HLs, the cost is not a priority but the proposed model tries to build fewer centers or centers in locations with less cost. Furthermore, perishable items such as food and medicine are crucial when a disaster occurs, so we have considered time window for perishable items which should be delivered at a specific time; otherwise, they will be useless. Transporting perishable items are not same as imperishable, because perishable RIs should be kept in special conditions such as suitable temperature, keeping out of the sun, etc. which make refrigerated vehicles very useful; on the other hand, imperishable RIs such as blankets or clothes do not need to be transported under special conditions, so we have defined

Table 1
A summary of related literature.

Ref	Pre-disaster	Post-disaster	Pre-positioning	Pre-disaster procurement	Post-disaster procurement	Multi-objective	Single objective	Perishable	imperishable
Baskaya et al. [10]	*		*	*	*		*		*
Bozorgi-Amiri et al. [12]	*	*	*	*	*	*			*
Khayal et al. [18]		*			*		*		*
Chakravarty [23]	*	*		*	*	*			*
Döyen et al. [24]	*	*	*				*		*
Mohammadi et al. [25]		*			*	*			*
Tofghi et al. [28]	*	*				*			*
Rezaei-Malek et al. [31]	*	*	*		*	*			*
Ozkapici et al. [32]		*					*		*
Rodríguez-Espíndola et al. [37]	*		*			*			*
Ahmadi et al. [39]		*					*		*
Condeixa et al. [40]	*	*					*		*
Manopiniwes and Irohara [42]	*	*			*	*			*
Rahmani et al. [43]	*	*	*				*		*
Yahyaei and Bozorgi-Amiri [44]	*	*					*		*
This study	*		*	*	*	*		*	*

different transportation methods for perishable and imperishable items.

We have made some assumptions to formulate our model:

- The demand in damaged areas, traveling time, traveling distance, loading-unloading time and costs are uncertain
- All demands in each demand center must be satisfied
- Each DP must be assigned to exactly one RC
- Each RI and transportation vehicle have a different volume
- Perishable and imperishable commodities should be transported by different vehicle
- Perishable items should be delivered before they decay otherwise, they'll be useless
- There is a limitation for maximum traveling distance each RI can travel
- RCs have infinite capacities for holding RIs
- Number of RCs that can be established is limited
- Loading and unloading time is equal for all vehicles

To sum up, in this paper, HL is viewed as a two-level network. The first level is the set of RCs and DPs are in the second level. The model determines required amount of RIs that should be pre-positioned at RCs, and number of vehicles to deliver RIs to DPs.

3.2. Mathematical formulation

A mixed-integer programming formulation is presented in this section to minimize the total distance traveled per relief commodity, inventory holding and transportation cost, maximum traveling time. Before submitting the optimization model, we provide the notations.

Sets

I	Set of RCs $i \in I$
j	Set of DPs $j \in J$
kp	Set of perishable RIs
kip	Set of imperishable RIs
k	Set of RIs $k = kp \cup kip$
op	Set of vehicles for perishable RIs
oip	Set of vehicles for imperishable RIs
o	Set of vehicles $o = op \cup oip$

Parameters

W	A large number
P	Maximum number of RCs to open
R	Maximum distance for a RI to travel
$\tilde{l}_o$	Expected vehicle loading time
$\tilde{u}_l$	Expected vehicle unloading time
u_k	Unit volume of item type k
$\tilde{h}_k$	Expected inventory holding cost for RI k (per unit)
$\tilde{c}_o$	Expected transportation cost for vehicle o (per unit)
$\tilde{d}_{jk}$	Expected demand for RI k in DP j
cp_o	The capacity of vehicle o
v_{ij}	Vulnerability factor between RC i and DP j
r_{ij}	Expected traveling time between RC i and DP j
$\tilde{t}_{ij}$	Expected travel time between RC i and DP j
$\tilde{E}s_i$	Expected establishing cost of each RC i
$\tilde{A}c_k$	Expected acquiring cost of RI k (per unit)
s_{kp}	Available time to deliver RI k before it decays

Variables

q_{ik}	Quantity of RI k held at RC i
x_{ijko}	Quantity of RI k sent to demand j from RC i by vehicle o
y_i	$\begin{cases} 1 & \text{if RC i is opened} \\ 0 & \text{otherwise} \end{cases}$
m_{ij}	$\begin{cases} 1 & \text{if DP j is assigned to RC i} \\ 0 & \text{otherwise} \end{cases}$

Pe_{kp}	$\begin{cases} 1 & \text{if RI kp decays} \\ 0 & \text{otherwise} \end{cases}$
Ri_{ik}	Number of remaining RI k in RC i after sending RIs to DPs
Nt_o	Number of vehicle o
T_{max}	Maximum traveling time

The following multi-objective MINLP can now be proposed:

$$\text{Min obj1} = \sum_o \sum_i \sum_j \sum_k x_{ijko} r_{ij} (1 + v_{ij}) \quad (1)$$

$$\text{Minobj2} = \sum_i \left(\tilde{E}s_i y_i \sum_k \tilde{A}c_k q_{ik} \right) + \sum_o \tilde{c}_o Nt_o + \sum_i \sum_k \tilde{h}_k Ri_{ik} \quad (2)$$

$$\text{Min obj3} = T_{max}, \quad (3)$$

$$\text{Min obj4} = \sum_{kp} Pe_{kp} \quad (4)$$

Subject to:

$$\sum_i \sum_o x_{ijko} = \tilde{d}_{jk}, \forall j, k \quad (5)$$

$$r_{ij} (1 + v_{ij}) m_{ij} \leq R, \forall i, j \quad (6)$$

$$\sum_j \sum_k x_{ijko} \leq \sum_k q_{ik}, \forall i, o \quad (7)$$

$$\sum_i y_i \leq P \quad (8)$$

$$\sum_i m_{ij} = 1, \forall j \quad (9)$$

$$\sum_j m_{ij} \leq W y_i, \forall i \quad (10)$$

$$x_{ijko} \leq W m_{ij}, \forall i, j, k, o \quad (11)$$

$$\sum_i q_{ik} \leq \left\{ \sum_j \tilde{d}_{jk} \right\} 1.01, \forall k \quad (12)$$

$$q_{ik} \leq W y_i, \forall i, k$$

$$T_{max} \geq m_{ij} \tilde{t}_{ij} (1 + v_{ij}), \forall i, j \quad (13)$$

$$\sum_j \sum_o x_{ijko} + Ri_{ik} \leq q_{ik}, \forall i, k \quad (14)$$

$$(t_{ij} + \tilde{l}_o + \tilde{u}_l) (1 - Pe_{kp}) m_{ij} \leq s_{kp}, \forall i, j, kp \quad (15)$$

$$Nt_o \geq \frac{\left(\sum_i \sum_j \sum_k x_{ijko} u_k \right)}{cp_o}, \forall o \quad (16)$$

$$x_{ijko} \geq 0, \forall i, j, k, o \quad (17)$$

$$q_{ik} \geq 0, \forall i, k \quad (18)$$

The objective function (1) minimizes distance traveled by RIs, the objective function (2) minimizes RC establishing cost, RI acquisition cost, vehicle purchasing cost, and total inventory holding cost. The objective function (3) minimizes the maximum traveling time from facility i to demand location j and objective function (4) minimizes the total quantity of perished items. Constraint (5) provides that demand at each DP should be satisfied. Constraint (6) demonstrates that RIs cannot travel more than R, and relief commodities sent to demand centers do not exceed respective inventory held at RC i by constraint (7). Constraint

(8) limits the RC centers that can be opened. Constraints (9), (10) and (11) provides that each DP i is assigned to a RC which is opened and RIs cannot be sent from an RC to a DP unless that demand location is assigned to that RC. Constraint (12) demonstrates that the total capacity of RC is 101% of total demand. Constraint (13) provides an upper bound for traveling time, and constraint (14) allows surplus items to be stored in RCs. Constraint (15) provides perishable item is decayed if it delivers after it expires time and constraint (16) ensure that volume limitation for vehicles should be fulfilled.

4. Solution methods

The proposed model is a non-linear multi-objective model. To deal with multi-objective function we used Lp-metric method which is described in next section. Furthermore, because of non-linearity, solving the model with exact methods for big scale problems is time-consuming. Thus, in this research, not only do we use an exact approach to solve our model, but we also use Grasshopper Optimization Algorithm (GOA) as a meta-heuristic approach to solving some problems in big scale. Table 2 demonstrates a review of solution methods used in literature before.

With respect to Table 2, most of the researchers preferred to use an exact approach to solve their model even though solving problems in large scale takes long time.

4.1. Multi-objective approach

There are various approaches in the literature to solve multi-objective optimization problems in HL. In this paper, we used Lp-metric technique that tries to minimize distances between objective functions without any primitive information from decision-maker. In other words, deviation of objective functions is minimized in Lp-metric technique [46].

First, optimal values for each objective function should be calculated separately. Then, anti-ideal values for objective functions should be determined (in our case, it means maximum values of all objective functions) and by placing obtained values in equation (19), the Lp model is minimized with model constraints. In this way, deviation from optimal values are minimized [47]. It should be noted that w_p represents the importance of each objective function and p shows emphasis on deviation.

$$Lp = \left\{ \sum_{j=1}^k w_j \left[\frac{f_j(x^{*j}) - f_j(x)}{f_j(x^{*j}) - f_j(\bar{x})} \right]^p \right\}^{\frac{1}{p}} \quad (19)$$

4.2. Grasshopper optimization algorithm

GOA was proposed by Saremi et al., in 2017. This algorithm is a meta-heuristic method which is inspired from grasshopper swarms'

behavior. GOA is a population-based algorithm that seeks to find optimal solutions Swarm intelligence techniques. GOA mimics grasshoppers' behavior and their movements for finding food sources. This algorithm can outperform other recent and well-known meta-heuristic algorithms, and it has been proven that GOA can solve real world problems with unknown search spaces Saremi et al. [48]. Nature-inspired algorithms logically divide searching process into two propensities namely, exploration and exploitation that naturally grasshoppers carry out. To be more specific, immature grasshoppers do exploration because they move slowly with small steps whereas mature grasshoppers have abrupt movements with big steps that can be considered as exploitation. As a result, with modelling grasshoppers' behavior we can create a powerful nature-inspired algorithm for optimizing problems Saremi et al. [48].

GOA has been developed for optimization over continuous spaces; consequently, for discrete spaces, initial solutions should be created continuously in order to exploration and exploitation operators work properly. Then, continuous variables should be converted to discrete ones. For example, to create binary variable y_i , a continuous vector between zero to one with P components (number of RCs that can be established at most) is created. Afterwards, components are sorted ascending and another vector that indicates order of indices is created. So, a random arrangement between 1 to P is obtained and because P RCs should be established at most, another random number (call it PP) between 1 to P is created and RCs establish from 1 to PP from the second vector. So, in each iteration, various RCs and different number of RCs are chosen randomly. Fig. 1 depicts how variable y_i is created when at most five RCs can be established.

Nonlinearity in a mathematical model increases solution time significantly. Fig. 2 demonstrates how CPU time increases exponentially when problem size increases. Therefore, solving the model with exact method is time-consuming. In this paper, we employed a new meta-heuristic approach so as to solve big scale problems in a reasonable time. In population-based algorithms, first, an initial population of solutions are created and, in each iteration, solutions improve with respect to objective function. Because all decision variables have values in the beginning and all equations are calculated based on given values, variables act as parameters that eliminates nonlinearity effect. In this model, second objective function and equation (15) are nonlinear, so, we created m_{ij} , Pe_{kp} , q_{ik} , y_i as grasshoppers' initial population-which is discussed in previous paragraph elaborately- and improved by GOA's operators in each iteration. For instance, y_i is created as a numerical vector and q_{ik} as numerical matrix; by multiplying these two we only have a vector that multiplies to a matrix. Accordingly, nonlinearity is eliminated in GOA unlike what we have in GAMS.

4.3. Tuning parameters and numerical example

In this section, we solve two test problems with the proposed model.

Table 2

A review of solution methods in several related studies.

Study	Solution method		
	Exact	Heuristic	Meta-heuristic
Baskaya et al. [10]	CPLEX		
Bozorgi-Amiri et al. [12]	Lp-metrics		
Khayal et al. [18]	CPLEX		
Döyen et al. [24]	CPLEX	Augmented Lagrangian	
Mohammadi et al. [25]	Optimization solver		Particle swarm optimization
Tofighi et al. [28]	Optimization solver		Differential evolution algorithm
Rezaei-Malek et al. [31]	Augmented e-constraint method		NSGA-II
Rodríguez-Espíndola et al. [37]	e-constrained method		
Ahmadi et al. [39]	CPLEX		Variable neighborhood search
Condeixa et al. [40]	CPLEX		
Rahmani et al. [43]		Lagrangian relaxation	
Yahyaee and Bozorgi-Amiri [44]	CPLEX		

A random continuous vector

RC	1	2	3	4	5
UNIF(0,1)	0.4	0.2	0.5	0.7	0.3



Sorted vector

RC	2	5	1	3	4
UNIF(0,1)	0.2	0.3	0.4	0.5	0.7



A random permutation of RCs

2	5	1	3	4
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Fig. 1. A schematic view of variable creation in GOA.

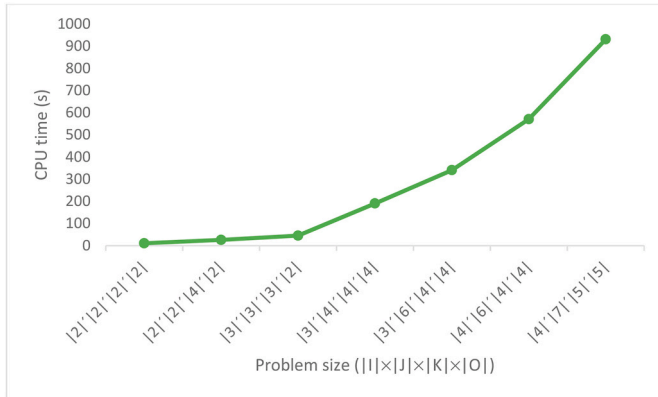


Fig. 2. CPU time vs problem size.

We tried to use appropriate input data for each parameter based on the test problem size. For example, to understand the proposed model better and because of the deterministic solution approach for the small and medium size problem, parameters are considered smaller than they are in the real world. In contrast, for large-scale problems, parameters are more realistic.

To evaluate the proposed model, a small and a medium-size problem are solved by GAMS software (BARON solver). In the first problem, we consider three potential places to establish RCs, but only two centers can be established and there are two DPs. There is only one perish and imperishable item needed in each DP and there is one type of vehicle for each item type. Fig. 3 depicts how each RC should be allocated to DP considering transportation methods (as mentioned before, there are different type of vehicles). Results for this problem with different weights for objective functions are shown in Fig. 4. It should be noted that we have considered three values for each uncertain parameter, namely, optimistic, pessimistic and most likely. Then mean of the uncertain parameter are calculated with the following formula Perry and Greig [45]:

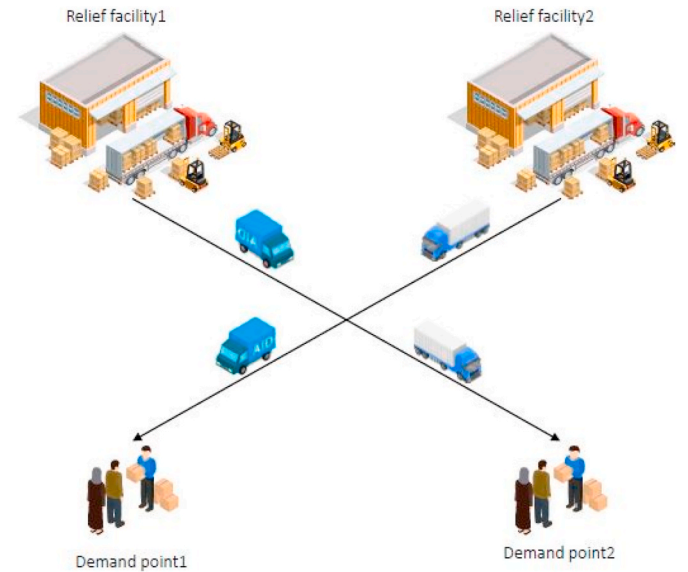


Fig. 3. An illustration of the first test problem result.

$$\mu = \frac{x_o + 4x_m + x_p}{6} \quad (20)$$

In the second test problem, we will consider five potential places to establish RCs, but only three of them could be chosen. There are six DPs with the need of two perishable and imperishable RI in each DP. In order to depict the results easier, we consider only two types of vehicle again. Fig. 5 illustrates a schematic view of second test problem and Pareto front relating to this problem is depicted in Fig. 6.

One of the most important stages in using meta-heuristic algorithms is to adjust parameters in order to improve the effectiveness of the algorithm. Population Size of Grasshopper (PSG), Maximum Iterations (MI) and Decrement factor (C) are three parameters that should be

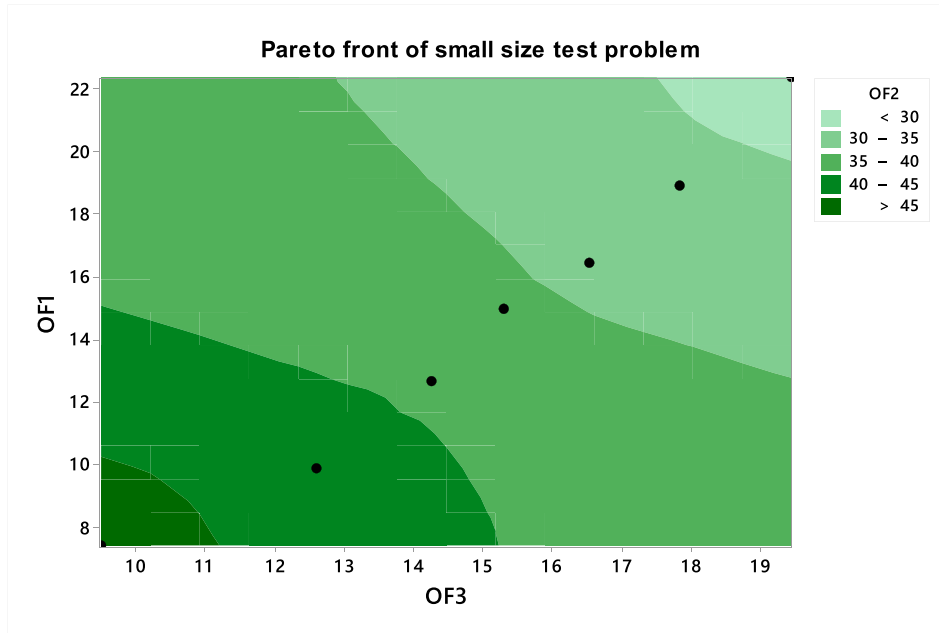


Fig. 4. Pareto front for the first test problem.

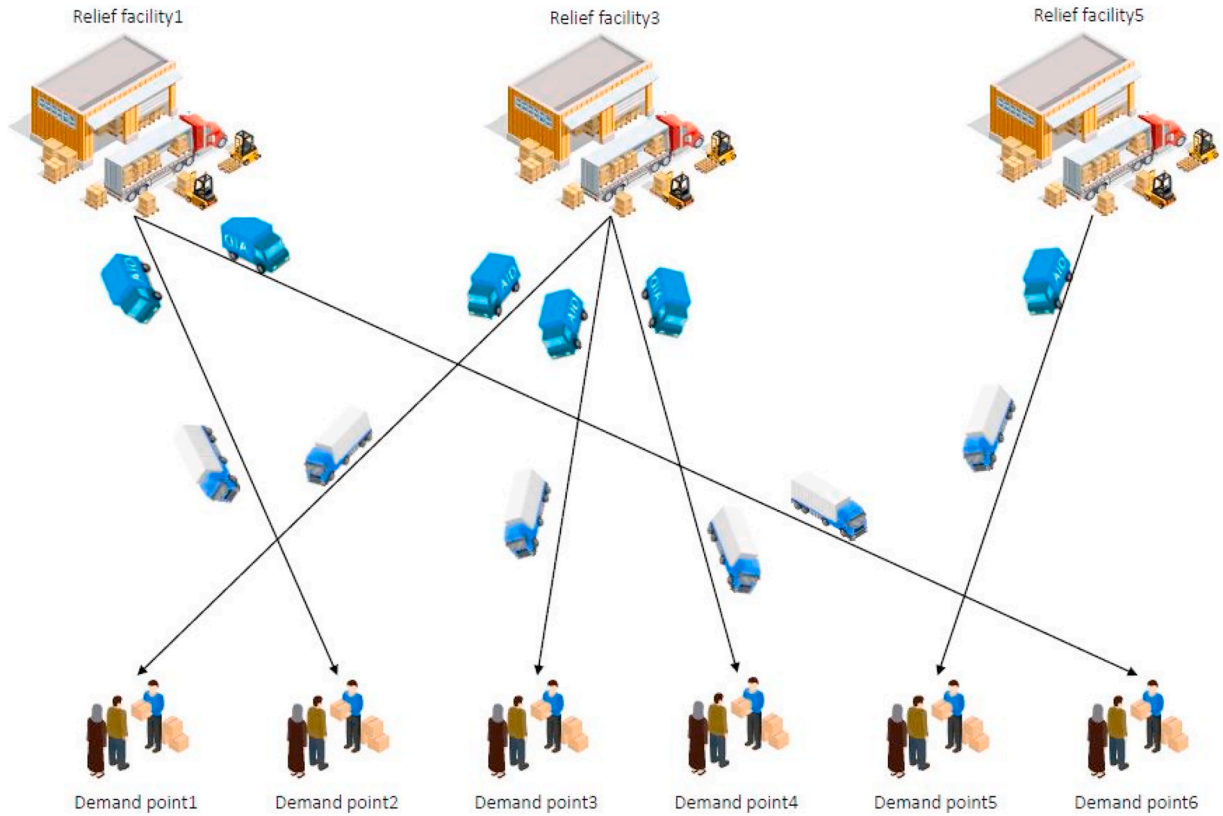


Fig. 5. An illustration of the second test problem result.

adjusted. In this paper, we used the Taguchi method with MINITAB software for tuning PSG, MI and C. Levels and factors considered to design experiment by Taguchi method are described in Table 3.

We used the L9 (3**2) design for tuning parameters. Normalized data from the Taguchi experiment are presented in Table 4. We normalized the results by Related Percentage Deviation (RPD) using

formula (20).

$$RPD = \frac{| \text{Method Sol} - \text{Best Sol} | * 100}{| \text{Best Sol} |} \quad (21)$$

Corresponding mean and SNR graph related to the Taguchi method is depicted in Fig. 7 and Fig. 8. According to results obtained from SNR and Mean graphs, minimum mean has occurred at third level for MI, at

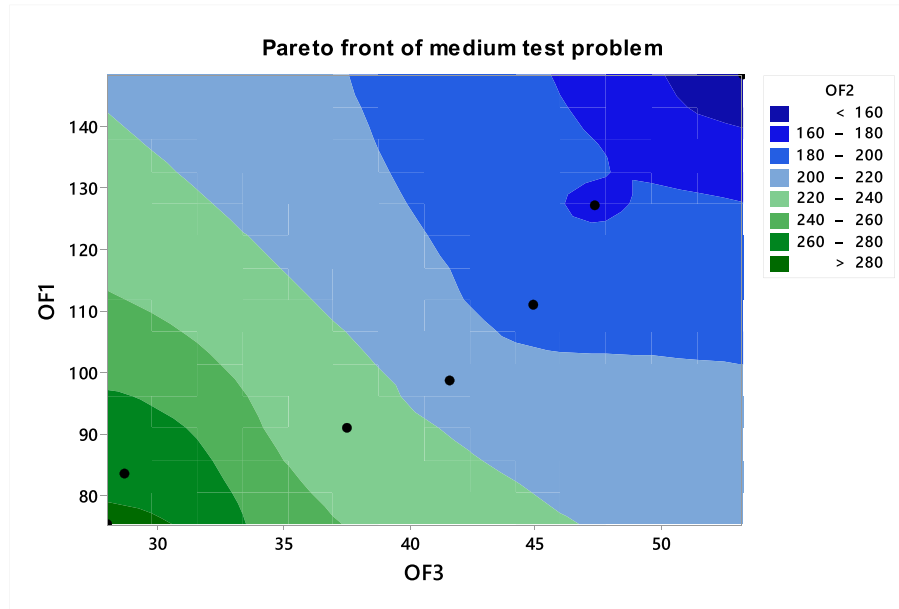


Fig. 6. Pareto front of second test problem.

Table 3
Levels and factors for GOA parameters.

level	Factors		
	MI	PSG	C
Low	500	75	0.7
Medium	750	100	0.8
High	1000	150	1

second level for PSG and at first level for C. Furthermore, in the SNR graph maximum signal to noise has occurred at the same level for these parameters. Finally, parameters for GOA can be adjusted, as shown in Table 5.

To evaluate and validate the proposed algorithm with adjusted parameters, nine test problems are solved by GAMS and MATLAB software. We used an AMD Corei5 laptop with 2.7 GHz and 8 GB RAM in MATLAB R2017b to solve test problems. First, four test problems are small and the rest of them are medium scale. Data required for test problems are created randomly by considering real-world conditions. Results are compared in Table 6. formula (22) is used for calculating gap rate.

$$\text{Gap Rate} = \frac{|\text{Best Sol} - \text{Test Sol}|}{|\text{Best Sol}|} * 100 \quad (22)$$

According to Table 5 and statistical hypothesis testing performed in appendix A, there is only a slight deviation between results obtained from GOA and GAMS so we can conclude that the proposed algorithm has proper efficiency with adjusted parameters.

4.4. Model validation

In this part, to validate the proposed model's function, we will change some parameters that affect objective functions as we expect. For example, parameters r_{ij} and t_{ij} indicate the distance and travel time between DPs and RCs, respectively. By increasing all distances and traveling times between RCs and DPs, first and third objective functions increase (traveling distance for RIs and maximum traveling time) but the second objective function remains unchanged. Increasing r_{ij} and t_{ij} should not influence the second objective function. It must be emphasized that there is limitation for establishing RCs. Fig. 9 illustrates the change in objective functions with different traveling time, traveling

Table 4
Normalized data by RPD.

Run	First OF	RPD	Second OF	RPD	Third OF	RPD
1	624.62	0.54	129.91	3.68	46.74	3.37
2	623.19	0.31	131.34	4.82	46.04	1.84
3	633.02	1.89	130.22	3.92	46.55	3.48
4	634.92	2.20	126.22	0.74	47.31	4.65
5	632.22	1.76	129.41	3.28	45.43	1.01
6	631.46	1.64	127.13	1.47	45.55	1.27
7	621.28	0.00	126.62	1.06	45.21	0.00
8	621.51	0.04	125.29	0	46.08	1.91
9	622.42	0.18	129.93	3.70	46.43	2.69

distances.

Other parameters that have a direct impact on objective functions are cost parameters such as inventory and RC establishment cost. By enhancing these parameters, the second objective function should increase and other objective functions are expected to stay unaltered. Fig. 10 shows the second objective function with different values related to cost. It is clear that by increasing cost parameters, the second objective function increases.

5. Case study

On March 19, 2019 until 29 April, a widespread flash flood occurred in 31 provinces of Iran and affected approximately 12 million people. It led to 70 deaths, four missing people, more than 1000 injuries, and more than \$2.2 billion (USD) financial damages. Furthermore, about 114,000 houses were damaged, and 65,000 houses were destroyed by the flood [49]. Although the flood affected more than 20 provinces, severity in several provinces such as Golestan, Khuzestan, Lorestan and Fars caused more damages to agricultural infrastructures. Fig. 11 delineates Iran's map which is extracted from (<https://commons.wikimedia.org/wiki/File:IranGolestan-SVG.svg?uselang=fa>) and most damaged areas are highlighted in blue. In the present research, we only implement our model to Golestan province that is marked with a red circle.

It should be noted that in section 6.2, we have already ratified the performance of the proposed meta-heuristic method and in this section, we only implement the proposed method for solving case study's model.

Golestan province is divided into fourteen counties and it has a

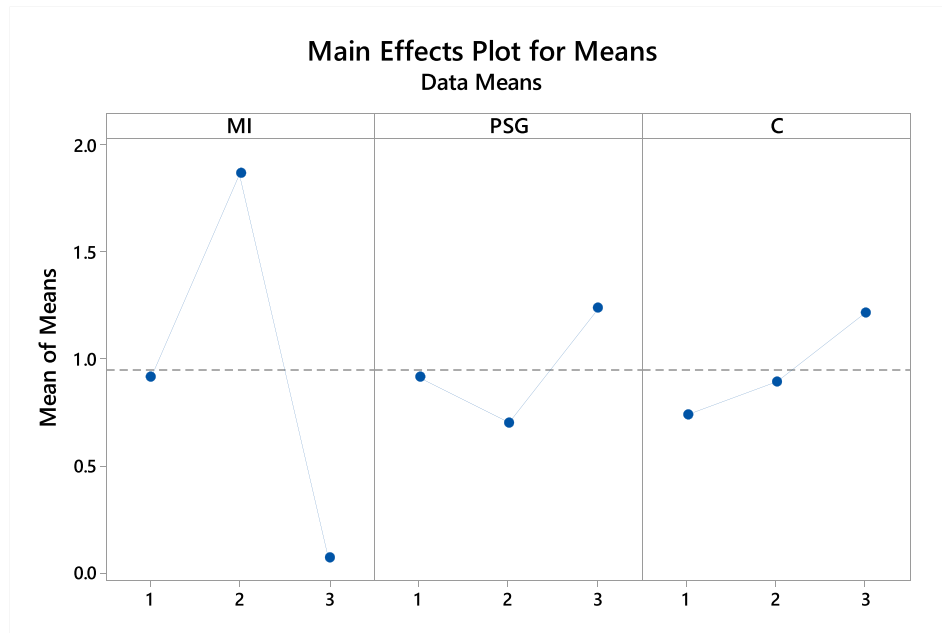


Fig. 7. Mean graph of Taguchi experiment.

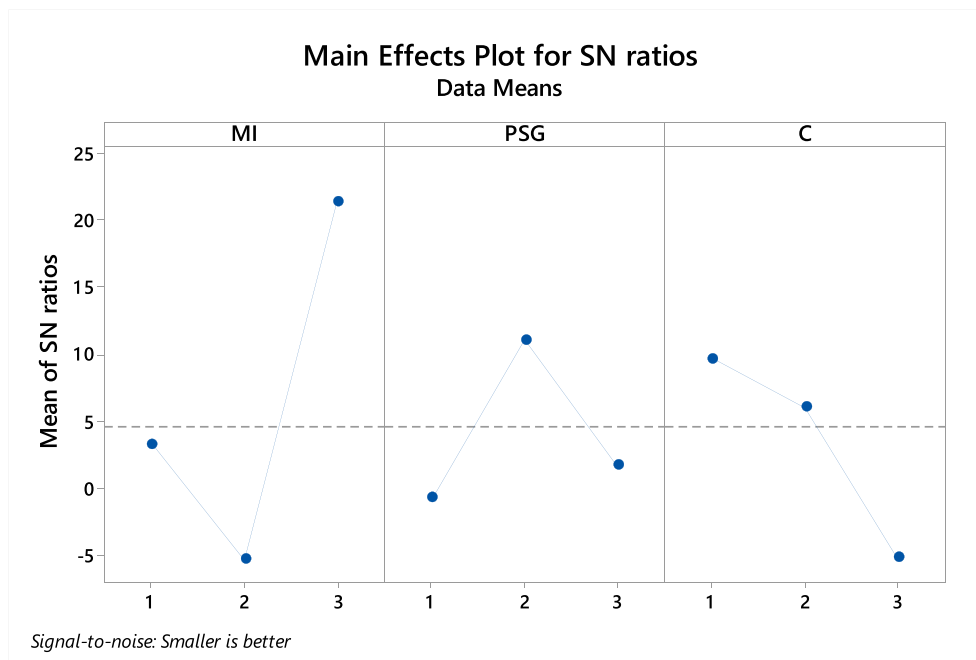


Fig. 8. SNR graph of Taguchi experiment.

Table 5
Tuned parameters value.

Factor	
MI	1000
PSG	100
C	0.8

population of 1,868,819 (2015) according to reports provided by the Statistical Center of Iran and an area of 20,380 km². Because of geographical and weather conditions, Golestan is a vulnerable region to be flooded and it has been exacerbated with human activities for converting river routes for urban development without planning for an

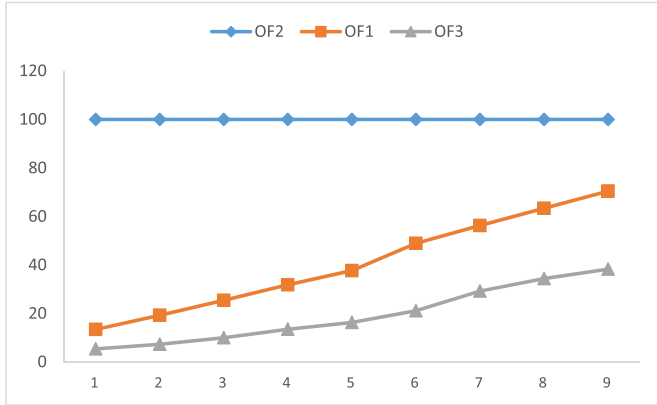
appropriate drainage system. According to reports provided by IRCS 62359 people had been affected and 15907 were moved to temporary shelters like stadiums (<http://en.rcs.ir/information-bulletin-12-iran-flood>).

For implementing the proposed model to this case, we considered fourteen potential places to establish FCs in each county but because of budget constraints, province's vastness and population, we set $P = 2$ (maximum number of facilities to open). It should be noted that flash flood affected mainly in several counties namely, Gonbad Kavus, Azadshahr, Ramian, Gorgan, Aq Qala, Kumus Depe, Bandar Torkaman and Kordkuy. Thus we considered the aforementioned counties in our case. Traveling distances are measured with google map's measure tool (<https://www.google.com/maps/@37.1111456,54.5312671,8.42z>)

Table 6

Solutions obtained by GOA and GAMS.

Test problem	GAMS				GOA				Gap(%)			
	1st obj (10 ³) meter	2nd obj (10 ⁴) dollar	3rd obj minute	4th obj	1st obj	2nd obj	3rd obj	4th obj	1st obj	2nd obj	3rd obj	4th obj
1	14.92	36.168	15.3	0	14.9236	36.169	15.39	0	1.61	0.26	0.248	0
2	32.8	94.305	17.5	0	32.8649	94.5	17.54	0	0.443	0.1	1.75	0
3	43.2	111.68	24.2	0	43.895	111.97	24.26	0	0.11	0.32	2.943	0
4	56.4	147.517	29.6	0	56.6498	147.65	30.12	0	0.43	0.225	0.95	0
5	64.64	187.352	36.7	4	64.71	187.952	37.78	4	1.3	0.36	0.9	0
6	83.54	203.278	39.85	5	83.899	203.736	40.23	5	0.3	0.2	1.8	0
7	98.45	212.851	41.6	7	99.741	213.62	41.98	7	0.47	0.47	4	0
8	112.63	235.612	43.9	10	112.971	236.08	44.7	10	1.61	0.26	0.248	0
9	135.78	257.692	45.15	12	136.417	258.92	46.96	12	0.443	0.1	1.75	0

**Fig. 9.** First and third objective functions value with a different traveling distance.

and then traveling time between RCs and DPs are estimated. Moreover, demand in each DP depends on the population of affected people living there, and it is supposed that every individual should receive a package of RIs. Packages contain ten different commodities; perishable item like food parcels and medicines, imperishable items such as blankets, groundsheets, etc. Data relating to this case study is provided in appendix B.

We also considered two types of vehicles, type I for perishable and type II for imperishable items. Results obtained from the proposed model are provided in Table 7 and the allocation of DPs to RCs are depicted in Fig. 12. In this case, the last objective function with different weights for Lp-metrics technique remains zero because distances between different points are not so long. Furthermore, input parameters for our case are provided in appendix B. Since, all type of RIs are supplied from one RC, we aggregated all quantities and introduced as q_{total} . If demand for RI type k , supplies from RCs in a way that for all RCs $t_{ij} + lo + ul \leq s_k$; none of RIs will perish. Thus, to minimize forth objective function, the model allocates DPs to RCS with respect to t_{ij} for each route and s_k for each RI so as to satisfy $t_{ij} + lo + ul \leq s_k$.

Moreover, in this case, distances between counties were not considerably long because cities and counties are very close together in north of Iran; so amount of perished RIs is zero whereas by implementing this model for other cases in bigger areas, t_{ij} increases and it may lead the forth objective function not to be zero. Related decision variables for this case study are demonstrated in Table 8.

6. Sensitive analysis

According to the proposed model, we will survey on the effects of different parameters by changing them and evaluate their impact on the model. Parameter P in our model, which represents the maximum number of facility centers to open, will be surveyed first. Although cost is not the first priority in the humanitarian supply chain but in the real world, there is always some budget limitation. If it were possible to

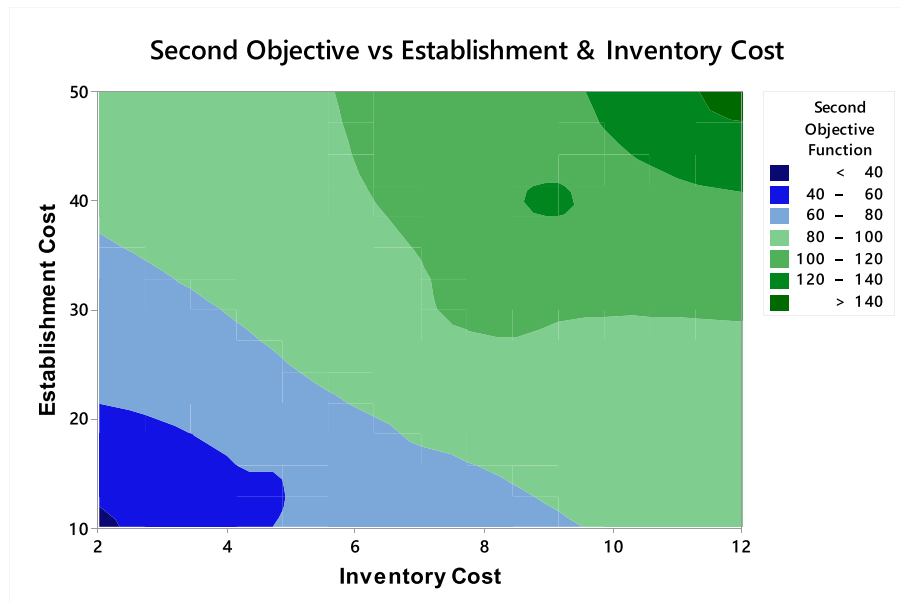
**Fig. 10.** Second objective function results with different cost parameters.



Fig. 11. Iran's map (affected regions are highlighted).

Table 7

Objective functions obtained by GOA algorithm.

1st OF	2nd OF (Rials)	3rd OF (hour)	4th OF
5227642.62	12429135570	1.215	0

establish RCs in all candidate places, traveling time and distance would decrease significantly, but by opening more RCs, the related cost would grow and it leads to increment in the second objective function. In other words, we can conclude that by ignoring cost issues, RCs would be established in all potential places. Fig. 13 shows alterations with different values for P .

Vulnerability factor is an uncontrollable parameter that depends on geographical conditions and severity of the disaster that may happen, that is determined by experts. More vulnerability factor indicates that if a disaster occurs the route between i and j are more susceptible to be harmed and vehicles should take a longer route. In other words, increasing severity of natural disasters leads to more damage to routes as well and as a result, RIs should be transported with alternative routes with longer distances. Fig. 14 illustrates objective functions with various vulnerable factor values.

We supposed that each RC has infinite capacity; therefore, shortcoming for imperishable commodities are unacceptable, but perishable items may decay before they deliver. Amount of perishable items depends on input parameters such as traveling time and expire date. Our last objective function tries to assign DPs to RCs in such a way that the number of perished items remain at the minimum level as we can see in our case study, by regarding gathered information and data, the amount of perished items is zero. The last objective function becomes more effective when distances become longer, for example, sending RIs from a capital city to a faraway township or sending international aids from different countries to an affected region.

Finally, by increasing only the distance between RCs and DPs, the

first objective function increases as we expect, but there are some points that the second objective function increases while the first objective function decreases. That is because the model tries to minimum distance traveled by RIs. By establishing RCs even with more cost, the model keeps the first objective function at the minimum level. Fig. 15 delineates how first and second objective behaves by enhancing distance. It should be noted that first and third objective function has the same behavior and by comparing second and third objective function, we obtain the same results.

7. Discussion and managerial insights

Preparedness before and response rapidly after the disaster is crucial for governments and without a good plan, any disaster may cause many casualties. After the occurrence of catastrophe, it is not the time for planning how to rescue or to evacuate affected people. In this research, we surveyed on both preparedness and response stage. Based on our findings, we can propose several useful managerial insights as follows: before the occurrence of any disaster, governments should keep themselves prepared by establishing RCs, especially in regions that are susceptible to natural disasters. In addition, the amount of RIs that may be needed should be determined with respect to the population of that area. Since perishable items like food parcels and medicines play a crucial role in rescuing affected people, governments should pay more attention to them. At the second stage, a distribution plan that assigns vehicles to DPs from different facilities should be provided and the number of vehicles should be calculated. However, in humanitarian relief chain, usually cost is not the main concern but in order to help governments to conquer budget constraints, we considered cost issues in our model. Moreover, based on sensitivity analysis, some parameters such as the severity of disaster has a direct impact on some decision variables and objective functions. Therefore, this research can help decision-makers to make the right decisions properly. To some up, this paper help decision-makers in all three conventional levels of decision-making, i.e., strategic for long planning horizon (pre-disaster), tactical and operational for real-time information (post-disaster).

8. Conclusion and future study

In this paper, we introduced a multi-objective MINLP mathematical model with several uncertain parameters. The model tries to minimize total distance RIs travel, total costs, maximum traveling time and perished RIs, determine inventory level in RCs, locate RCs, and allocate DPs to them. Our model also considers different type of vehicles, perishable and imperishable RIs, loading and unloading time on vehicles and provides a distribution plan for delivering RIs. Moreover, the effect of different costs and traveling distance for RIs on establishing RCs and allocating DPs has been surveyed by performing a sensitivity analysis on some parameters.

To prove the applicability of the proposed model some random test problems have been solved in different sizes. And for test problems in large scale, we used GOA as a meta-heuristic method and numerical results have proved the efficiency of the proposed solution method. Finally, the proposed model is applied to a real case study in Golestan province in Iran.

For future study, this model can be extended in some directions: the proposed model can be developed with different relief operations like rescue vehicle routing problem, evacuation affected people and considering different scenarios. Furthermore, objective functions such as maximum coverage, maximum satisfaction level, and maximum reliability can be added to the model.



Fig. 12. Established RCs and allocations in Golestan.

Table 8
Decision variables related to case study.

Decision variables		
q_{total}	Gorgan	4134060
	Gonbad Kavus	1016775
Nt	vehicle I (for imperishable RIs)	205
	vehicle II(for perishable RIs)	90

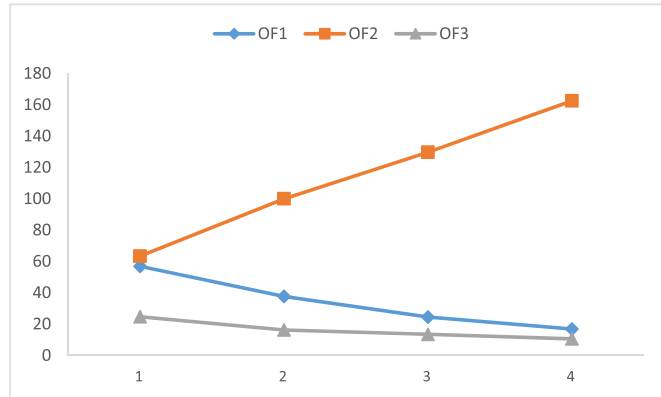


Fig. 13. First, second and third objective functions variation with a different value for P.

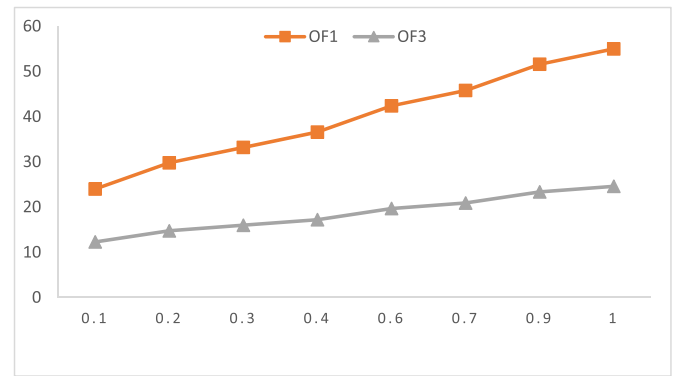


Fig. 14. Vulnerability factor's effect on objective functions one and three.

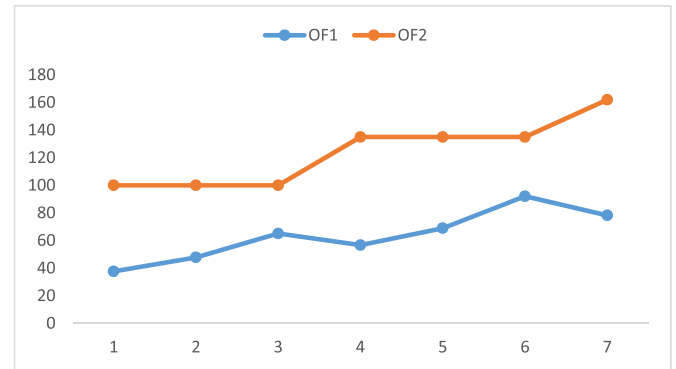


Fig. 15. First & second OF with different distance value.

Appendix A

In this part, statistical hypothesis testing about equality of objective functions value obtained by GAMS and GOA algorithm has been surveyed. Thus, a medium-size test problem has been solved by the GOA algorithm 25 times. Normality test for these data has been performed with Minitab software. Conforming to Kolmogorov-Smirnov test, which is plotted in A.1, A.2 and A.3 for different objective functions and regarding p-value, which is more than 0.05, presuming normal distribution for data obtained by GOA algorithm, is rational.

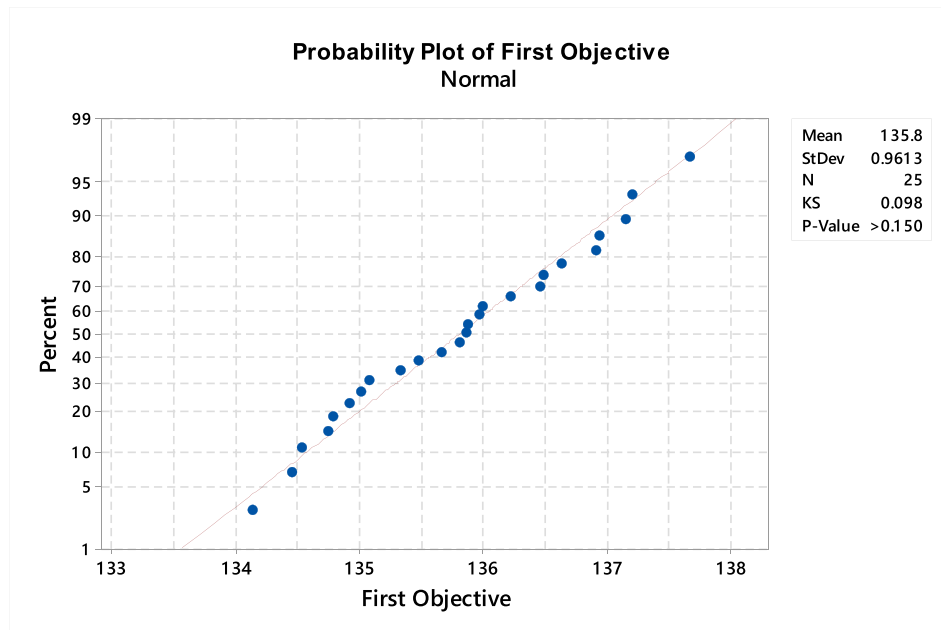


Fig. A.1. Kolmogorov-Smirnov test for the first objective function.

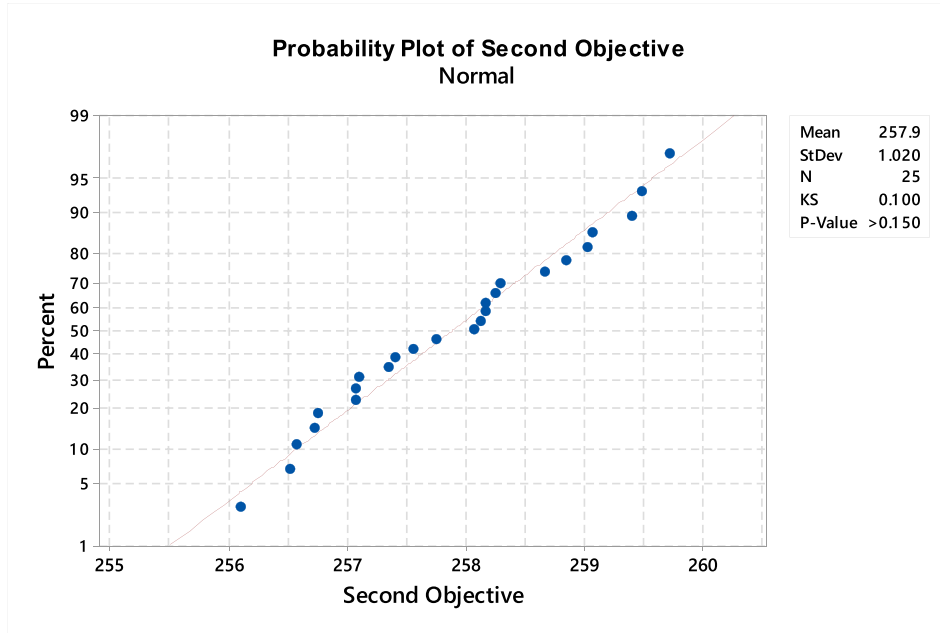


Fig. A.2. Kolmogorov-Smirnov test for the second objective function.

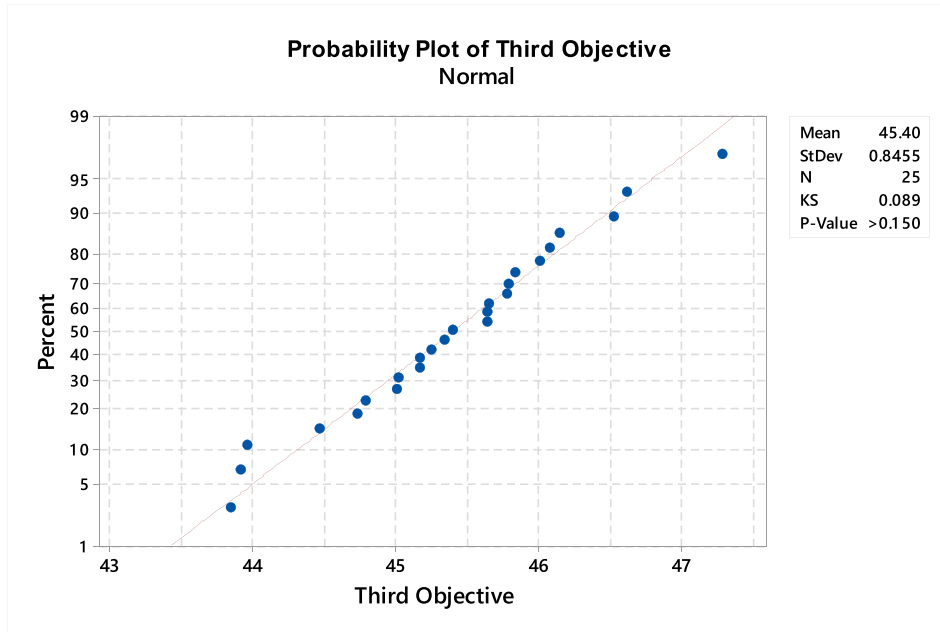


Fig. A.3. Kolmogorov-Smirnov test for third objective function.

According to Montgomery [50] for hypothesis test, we can use the following calculation:

$$\begin{cases} H_0: Z_{GOA} = Z_{GAMS} \\ H_1: Z_{GOA} > Z_{GAMS} \end{cases} \quad (A.1)$$

A statistic for this statistical hypothesis test is calculated by following equation:

$$t_0 = \frac{\bar{Z}_{GOA} - Z_{GAMS}}{\frac{S}{\sqrt{n}}} \quad (A.2)$$

$$S = \sqrt{\frac{\sum_{i=1}^n (Z_{(GOA)_i} - \bar{Z}_{GOA})^2}{n-1}} \quad (A.3)$$

In addition, because all of objective functions are minimizing, the acceptance region is $(-\infty, t_{\alpha, n-1}]$. Table A1 illustrates results obtained by calculating aforementioned equations.

Table A.1
Numerical results of the statistical hypothesis test.

First objective function	$\bar{Z}_{GOA}$	135.807
	Z_{GAMS}	135.78
	t_0	0.1379
	Acceptance region	$(-\infty, 1.711)$
Second objective function	$\bar{Z}_{GOA}$	257.8848
	Z_{GAMS}	257.692
	t_0	0.94480
	Acceptance region	$(-\infty, 1.711)$
Third objective function	$\bar{Z}_{GOA}$	45.4349
	Z_{GAMS}	45.12
	t_0	1.635
	Acceptance region	$(-\infty, 1.711)$

Based on results demonstrated in [table A2](#), in the significance level of 0.05, all sample statistics are in the acceptance region and null hypothesis cannot be rejected. Thus, there is not a considerable difference between the results obtained with these solution methods. Since values for the last objective function had no deviation with both exact and meta-heuristic method, performing a statistical hypothesis test was not necessary.

Appendix B

Table B.1
Estimated total demand in each county.

Azadshahr	102395
Ramian	74405
Gorgan	708227
Kordkuy	79821
Torkaman	90955
Gomishan	64594
Aqqala	73178
Gonbad Kavus	236606

Table B.2
Vulnerability factor of routes between counties.

	Maraveh Tappeh	Kalaleh	Galikash	Minudasht	Azadshahr	Ramian	Aliabad	Gorgan	Kordkuy	Bandar Gaz	Torkaman	Gomishan	Aqqala	Gonbad Kavus
Maraveh Tappeh		0.82	0.62	0.05	0.57	0.18	0.71	0.68	0.61	0.67	0.87	0.58	0.75	0.17
Kalaleh	0.82		0.36	0.22	0.39	0.40	0.71	0.65	0.55	0.47	0.58	0.18	0.29	0.96
Galikash	0.62	0.36		0.22	0.79	0.11	0.77	0.52	0.96	0.74	0.25	0.41	0.67	0.51
Minudasht	0.05	0.22	0.22		0.80	0.93	0.98	0.71	0.55	0.54	0.28	0.90	0.72	0.93
Azadshahr	0.57	0.39	0.79	0.80		0.83	0.78	0.95	0.80	0.26	0.32	0.62	0.94	0.39
Ramian	0.18	0.40	0.11	0.93	0.83		0.05	0.74	0.99	0.22	0.37	0.09	0.96	0.28
Aliabad	0.71	0.71	0.77	0.98	0.78	0.05		0.56	1.00	0.57	0.92	0.35	0.84	0.47
Gorgan	0.68	0.65	0.52	0.71	0.95	0.74	0.56		0.74	0.05	0.90	0.66	0.46	0.63
Kordkuy	0.61	0.55	0.96	0.55	0.80	0.99	1.00	0.74		0.12	0.92	0.94	0.17	0.97
Bandar Gaz	0.67	0.47	0.74	0.54	0.26	0.22	0.57	0.05	0.12		0.12	0.08	0.51	0.08
Torkaman	0.87	0.58	0.25	0.28	0.32	0.37	0.92	0.90	0.92	0.12		0.46	0.32	0.93
Gomishan	0.58	0.18	0.41	0.90	0.62	0.09	0.35	0.66	0.94	0.08	0.46		0.97	0.33
Aqqala	0.75	0.29	0.67	0.72	0.94	0.96	0.84	0.46	0.17	0.51	0.32	0.97		0.35
Gonbad Kavus	0.17	0.96	0.51	0.93	0.39	0.28	0.47	0.63	0.97	0.08	0.93	0.33	0.35	

Table B.3
Distance between counties in kilometer

	Maraveh Tappeh	Kalaleh	Galikash	Minudasht	Azadshahr	Ramian	Aliabad	Gorgan	Kordkuy	Bandar Gaz	Torkaman	Gomishan	Aqqala	Gonbad Kavus
Maraveh Tappeh		66.73	83.14	90.84	114.84	122.08	145.72	178.74	204.3	219.8	200	190	165.6	100
Kalaleh	66.73		12.19	19.38	43.56	50.21	76.14	110.59	139.13	152.9	136.48	128.49	100.46	35.15
Galikash	83.14	12.19		7.18	30.92	23.75	39.9	100	79.7	143	127	76	56.5	23
Minudasht	90.84	19.38	7.18		23.6	19.4	57.7	93.9	121.45	136	113	116.3	85	17.4
Azadshahr	114.84	43.56	30.92	23.6		8.3	33.6	69.6	62	114.8	99.5	98	64.78	17.11
Ramian	122.08	50.21	23.75	19.4	8.3		27.55	65.1	94.7	109.8	95.2	94.8	60.7	23.75
Aliabad	145.72	76.14	39.9	57.7	33.6	27.55		36.6	67.19	82.1	69.5	71.9	37.8	45
Gorgan	178.74	110.59	100	93.9	69.6	65.1	36.6		29.2	43.6	32.6	40.38	17.39	77.77
Kordkuy	204.3	139.13	79.7	121.45	62	94.7	67.19	29.2		13.57	11.95	30.6	38.5	106.6

(continued on next page)

Table B.3 (continued)

	Maraveh Tappeh	Kalaleh	Galikash	Minudasht	Azadshahr	Ramian	Aliabad	Gorgan	Kordkuy	Bandar Gaz	Torkaman	Gomishan	Aqqala	Gonbad Kavus
Bandar Gaz	219.8	152.9	143	136	114.8	109.8	82.1	43.6	13.57		17.08	53.46	54.7	119.85
Torkaman	200	136.48	127	113	99.5	95.2	69.5	32.6	11.95	17.08		17.67	34.44	103.72
Gomishan	190	128.49	76	116.3	98	94.8	71.9	40.38	30.6	53.46	17.67		33.9	98.93
Aqqala	165.6	100.46	56.5	85	64.78	60.7	37.8	17.39	38.5	54.7	34.44	33.9		67.77
Gonbad Kavus	100	35.15	23	17.4	17.11	23.75	45	77.77	106.6	119.85	103.72	98.93	67.77	

Table B.4

Traveling time between counties in hour.

	Maraveh Tappeh	Kalaleh	Galikash	Minudasht	Azadshahr	Ramian	Aliabad	Gorgan	Kordkuy	Bandar Gaz	Torkaman	Gomishan	Aqqala	Gonbad Kavus
Maraveh Tappeh	0.00	1.40	1.66	1.80	2.30	2.40	2.90	3.60	4.00	4.40	4.00	3.80	3.30	2.00
Kalaleh	1.33	0.00	0.25	0.38	0.87	1.00	1.50	2.20	2.80	3.00	2.70	2.60	2.00	0.70
Galikash	1.66	0.24	0.00	0.14	0.60	0.50	0.80	2.00	1.60	2.80	2.50	1.50	1.13	0.46
Minudasht	1.82	0.39	0.14	0.00	0.47	0.40	1.20	1.80	2.40	2.70	1.37	2.30	1.70	0.35
Azadshahr	2.30	0.87	0.62	0.47	0.00	0.17	0.67	1.39	1.24	2.30	1.99	1.96	1.30	0.34
Ramian	2.44	1.00	0.48	0.39	0.17	0.00	0.55	1.30	1.89	2.20	1.90	1.90	1.21	0.48
Aliabad	2.91	1.52	0.80	1.15	0.67	0.55	0.00	0.73	1.34	1.64	1.39	1.44	0.76	0.90
Gorgan	3.57	2.21	2.00	1.88	1.39	1.30	0.73	0.00	0.58	0.87	0.65	0.81	0.35	1.56
Kordkuy	4.09	2.78	1.59	2.43	1.24	1.89	1.34	0.58	0.00	0.27	0.24	0.61	0.77	2.13
Bandar Gaz	4.40	3.06	2.86	2.72	2.30	2.20	1.64	0.87	0.27	0.00	0.34	1.07	1.09	2.40
Torkaman	4.00	2.73	2.54	1.37	1.99	1.90	1.39	0.65	0.24	0.34	0.00	0.35	0.69	2.07
Gomishan	3.80	2.57	1.52	2.33	1.96	1.90	1.44	0.81	0.61	1.07	0.35	0.00	0.68	1.98
Aqqala	3.31	2.01	1.13	1.70	1.30	1.21	0.76	0.35	0.77	1.09	0.69	0.68	0.00	1.36
Gonbad Kavus	2.00	0.70	0.46	0.35	0.34	0.48	0.90	1.56	2.13	2.40	2.07	1.98	1.36	0.00

Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seps.2020.100933>.

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